Labs

Questions

1. What is observability, and why is it important in microservices architectures?
2. Explain the distinct role of metrics, logs, and traces in observability.
3. Signals, sources, agents, destinations are the components of observability architectures. Describe their role and mutual interactions.
4. What is defined as telemetry in the context of observability? In this context, describe the role of OpenTelemetry and its advantages over previous approaches.
5. What is the *Business Logic to Instrumentation* ratio? Describe the key observability-related costs.
6. Java provides a form of automatic, zero-code instrumentation. Describe it, and its advantages compared to manual alternatives.
7. What is Prometheus, and how does it facilitate metrics collection in applications? Which is its primary approach to metrics collection?
8. Describe the process of exposing application metrics using both Spring Boot Actuator and the Prometheus endpoint.
9. What is the ELK stack? Describe its architecture and key components.
10. Describe distributed tracing, traces, spans.
11. What is OpenTelemetry, how does its collector works?